

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

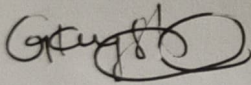
CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

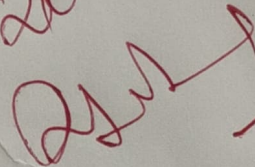
Code of Conduct:

I G.K. Vyshnavi ¹¹⁹²⁴²⁵⁰⁶⁹ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student



17
20



Assessment 1: TCP Three-Way Handshake in Wireshark

Aim

To capture TCP packets during connection setup and identify the **SYN**, **SYN-ACK**, and **ACK** packets.

Procedure

1. Open Wireshark and select the active **Wi-Fi** interface.
2. Start packet capture.
3. Open Command Prompt and execute:

```
curl https://example.com
```

4. Stop the capture.
5. Apply the filter:

```
tcp.flags.syn == 1
```

6. Identify the **SYN** and **SYN-ACK** packets.
7. Select the corresponding TCP conversation to identify the final **ACK** packet.
8. Observe the TCP flags in the packet details.

Output / Observation

Packet	TCP Flag	Purpose
1	SYN	Client requests a TCP connection
2	SYN, ACK	Server accepts the request and sends its own SYN
3	ACK	Client acknowledges the server

Result

432 41.19451600 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 306 GET /MFewTzBMEsW5TAJ8gUrDgKCGUABBR7tU9uFqgVQhJXKZYXCNXmVR5ngQLOBEK1z6uBQf54q8p74K1f9AwplQCEBc9.. 74 80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
433 41.19451600 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 1374 80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=1300 [TCP PDU reassembled in 434]
434 41.19451600 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. OCSP 1049 Response
435 41.19451600 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 1049 [TCP Retransmission] 80 - 62204 [PSH, ACK] Seq=1301 Ack=233 Win=131072 Len=975 [TCP PDU reassembled in 434]
436 41.238512900 2409:40f4:40d4:b73e.. 2606:4700:83b2:c460.. TCP 86 62204 - 80 [ACK] Seq=233 Ack=2276 Win=65280 Len=0 SLE=1301 SRE=2276

Frame 428: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface
Ethernet II, Src: Intel_f7:1b:15 (98:af:65:f7:1b:15), Dst: d6:b6:5e:24:33:12 (d6:b6:5e:24:33:12), Dst: 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 86 62204 - 80 [ACK] Seq=1 Ack=233 Win=131072 Len=0
Internet Protocol Version 6, Src: 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 86 62204 - 80 [ACK] Seq=1 Ack=233 Win=131072 Len=0
Transmission Control Protocol, Src Port: 62204, Dst Port: 80, Seq: 0, Len: 0

0000 d6 b6 5e 24 33 12 98 af 65 f7 1b 15 86 dd 60 0f ..53..
0010 c9 d6 00 20 06 3f 24 09 40 f4 40 d4 b7 3e 55 9275..
0020 3b 4d 62 78 a1 f1 26 06 47 00 83 b2 c4 60 07 c6&..
0030 8d c5 1e dc 25 e4 f2 fc 00 50 c9 91 3c 79 00 00P..
0040 00 00 00 02 ff ff 01 e5 00 00 02 04 05 01 01 03 ..P..
0050 03 08 01 01 04 02

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No.	Time	Source	Destination	Protocol	Length	Info
428	41.099514600	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	TCP	86	62204 - 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
429	41.158866900	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	86	80 - 62204 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1300 SACK_PERM WS=8192
430	41.159126000	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	TCP	74	62204 - 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
431	41.159471900	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	HTTP	306	GET /MFewTzBMEsW5TAJ8gUrDgKCGUABBR7tU9uFqgVQhJXKZYXCNXmVR5ngQLOBEK1z6uBQf54q8p74K1f9AwplQCEBc9.. 74 80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
432	41.194516900	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	74	80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
433	41.238406300	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	1374	80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=1300 [TCP PDU reassembled in 434]
434	41.238408000	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	OCSP	1049	Response
435	41.238408900	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	1049	[TCP Retransmission] 80 - 62204 [PSH, ACK] Seq=1301 Ack=233 Win=131072 Len=975 [TCP PDU reassembled in 434]
436	41.238512900	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	TCP	86	62204 - 80 [ACK] Seq=233 Ack=2276 Win=65280 Len=0 SLE=1301 SRE=2276

Frame 429: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface
Ethernet II, Src: d6:b6:5e:24:33:12 (d6:b6:5e:24:33:12), Dst: Intel_f7:1b:15 (98:af:65:f7:1b:15), Dst: 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 86 80 - 62204 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1300 SACK_PERM WS=8192
Internet Protocol Version 6, Src: 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 86 80 - 62204 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1300 SACK_PERM WS=8192
Transmission Control Protocol, Src Port: 80, Dst Port: 62204, Seq: 0, Ack: 1, Len: 0

0000 98 af 65 f7 1b 15 d6 b6 5e 24 33 12 86 dd 60 0f ..53..
0010 b8 fe 00 20 06 38 26 06 47 00 83 b2 c4 60 07 c688..
0020 8d c5 1e dc 25 e4 24 09 40 f4 40 d4 b7 3e 55 92&..
0030 3b 4d 62 78 a1 f1 00 50 f2 fc fb 0b 49 5c c9 91P..
0040 3c 7a 80 12 ff ff 2f 6f 00 00 02 04 05 14 01 01 ..P..
0050 04 02 01 03 03 0d

*Wi-Fi
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No.	Time	Source	Destination	Protocol	Length	Info
428	41.099514600	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	TCP	86	62204 - 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
429	41.158866900	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	86	80 - 62204 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1300 SACK_PERM WS=8192
430	41.159126000	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	TCP	74	62204 - 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
431	41.159471900	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	HTTP	306	GET /MFewTzBMEsW5TAJ8gUrDgKCGUABBR7tU9uFqgVQhJXKZYXCNXmVR5ngQLOBEK1z6uBQf54q8p74K1f9AwplQCEBc9.. 74 80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
432	41.194516900	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	74	80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
433	41.238406300	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	1374	80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=1300 [TCP PDU reassembled in 434]
434	41.238408000	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	OCSP	1049	Response
435	41.238408900	2606:4700:83b2:c460..	2409:40f4:40d4:b73e..	TCP	1049	[TCP Retransmission] 80 - 62204 [PSH, ACK] Seq=1301 Ack=233 Win=131072 Len=975 [TCP PDU reassembled in 434]
436	41.238512900	2409:40f4:40d4:b73e..	2606:4700:83b2:c460..	TCP	86	62204 - 80 [ACK] Seq=233 Ack=2276 Win=65280 Len=0 SLE=1301 SRE=2276

Frame 430: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
Ethernet II, Src: Intel_f7:1b:15 (98:af:65:f7:1b:15), Dst: d6:b6:5e:24:33:12 (d6:b6:5e:24:33:12), Dst: 2606:4700:83b2:c460.. 2409:40f4:40d4:b73e.. TCP 74 80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
Internet Protocol Version 6, Src: 2409:40f4:40d4:b73e.. 2606:4700:83b2:c460.. TCP 74 80 - 62204 [ACK] Seq=1 Ack=233 Win=131072 Len=0
Transmission Control Protocol, Src Port: 62204, Dst Port: 80, Seq: 1, Ack: 1, Len: 0

0000 d6 b6 5e 24 33 12 98 af 65 f7 1b 15 86 dd 60 0b ..53..
0010 c9 d6 00 14 06 3f 24 09 40 f4 40 d4 b7 3e 55 9275..
0020 3b 4d 62 78 a1 f1 26 06 47 00 83 b2 c4 60 07 c6&..
0030 8d c5 1e dc 25 e4 f2 fc 00 50 c9 91 3c 7a fb 0bP..
0040 49 6d 50 10 00 ff 01 d9 00 00 ..P..

Assessment 2: DNS Query Using UDP in Wireshark

Aim

To capture a DNS query using UDP in Wireshark and compare the UDP header with the TCP header.

Procedure

1. Open Wireshark and select Wi-Fi.
2. Start packet capture.
3. Open Command Prompt.
4. Enter:

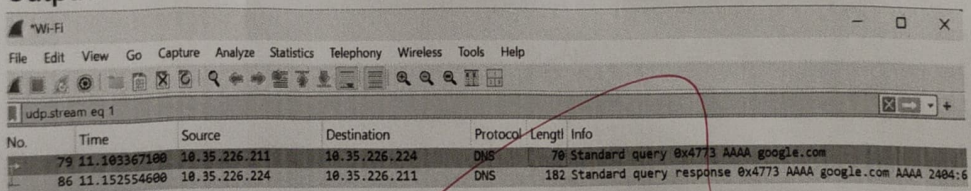
```
nslookup google.com
```

5. Stop the capture in Wireshark.
6. Apply the filter:

```
dns && udp
```

7. Select the DNS query packet.
8. Expand User Datagram Protocol and observe the UDP header.

Output:



No.	Time	Source	Destination	Protocol	Length	Info
79	11.103367100	10.35.226.211	10.35.226.224	DNS	70	Standard query 0x4773 AAAA google.com
86	11.152554600	10.35.226.224	10.35.226.211	DNS	182	Standard query response 0x4773 AAAA google.com AAAA 2404:6

```
> Frame 79: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface 0
> Ethernet II, Src: Intel_f7:1b:15 (98:af:65:f7:1b:15), Dst: c2:a8:5e:00:00:00
> Internet Protocol Version 4, Src: 10.35.226.211, Dst: 10.35.226.224
> User Datagram Protocol, Src Port: 63344, Dst Port: 53
> Domain Name System (query)
    0000 c2 a8 97 a5 fa 74 98 af 65 f7 1b 15 00 00 45 00 .....t..e
    0010 00 38 01 20 00 00 00 11 00 00 0a 23 e2 d3 0a 23 .8.....
    0020 e2 e0 f7 70 00 35 00 24 da 2f 47 73 01 00 00 01 ...p5$..
    0030 00 00 00 00 00 00 06 67 6f 6f 67 6c 65 03 63 6f .....gα
    0040 6d 00 00 1c 00 01 m.....
```

Assessment 3: TCP Retransmission vs UDP

Aim

To simulate packet loss and observe how TCP retransmits lost packets, while UDP does not retransmit them.

Procedure

Part A — Capture TCP retransmission

1. Open Wireshark.
2. Select Wi-Fi and start capturing.
3. Open Command Prompt.
4. Generate TCP traffic:

```
curl https://example.com
```

5. While traffic is being generated, briefly turn Wi-Fi OFF and ON.
6. Generate the traffic again if necessary.
7. Stop Wireshark.
8. Apply this filter:

```
tcp.analysis.retransmission
```

Wireshark may display:

[TCP Retransmission]

This indicates that TCP retransmitted a packet.

Part B — Observe UDP

Generate DNS traffic:

```
nslookup google.com
```

Use the filter:

```
dns && udp
```

You will see:

DNS Query

DNS Response

UDP itself does not show a TCP-style retransmission when a packet is lost.

Output:

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File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

No.	Time	Source	Destination	Protocol	Length	Info
11	0.000000	10.35.226.211	10.35.226.224	DNS	86	Standard query 0x0001 PTR 224.226.35.10.in-addr.arpa
12	0.000000	10.35.226.211	10.35.226.224	DNS	86	Standard query response 0x0001 No such name PTR 224.226.35.10.in-addr.arpa
13	0.000000	10.35.226.211	10.35.226.224	DNS	70	Standard query 0x0002 A google.com
14	0.000000	10.35.226.211	10.35.226.224	DNS	166	Standard query response 0x0002 A google.com A 192.178.158.138 A 192.178.158.139 A 192.178.158.139
15	0.000000	10.35.226.211	10.35.226.224	DNS	70	Standard query 0x0003 AAAA google.com
16	0.000000	10.35.226.211	10.35.226.224	DNS	182	Standard query response 0x0003 AAAA google.com AAAA 2404:6800:4013:813::65 AAAA 2404:6800:4013:813::65

Frame 160: Packet, 256 bytes on wire (2048 bits), 256 bytes captured (2048 bits) on interface 0	0000	90 af 65 f7 1b 15 c2 a8 97 a5 fa 74 81 00 00 00 00	C.....
Ethernet II, Src: Intel_7:7b:15 (98:af:65:f7:1b:15), Dst: Intel_7:7b:15 (98:af:65:f7:1b:15)	0010	06 d6 60 06 2a 43 00 c6 06 78 20 01 48 60 48 47	H.....
Internet Protocol Version 4, Src: 10.35.226.211, Dst: 10.35.226.224	0020	04 00 00 00 00 00 00 00 00 00 24 09 04 fa 44 8d	S.....
User Datagram Protocol, Src Port: 53098, Dst Port: 53	0030	c3 06 bc b6 46 15 a9 83 00 b4 81 b0 c6 b2 20 6c	F.....
Domain Name System (query), QID: 0, QType: A	0040	09 81 d6 84 44 11 50 10 04 12 8a f1 00 00 17 03	D.....
Domain Name System (query), QID: 0, QType: AAAA	0050	03 00 0d ff bb ac 67 da c2 f6 2a 18 3c 40 bc 45	S.....
Domain Name System (query), QID: 0, QType: PTR	0060	a9 a1 b8 b2 26 c0 08 85 82 78 7a a9 c3 34 97 43	S.....
Domain Name System (query), QID: 0, QType: PTR	0070	34 2a 49 92 47 f8 00 37 8a 99 82 76 54 40 64 53	S.....
Domain Name System (query), QID: 0, QType: PTR	0080	67 32 53 72 6a c2 c0 bd 39 fa 20 0d 1f 23 64 53	S.....
Domain Name System (query), QID: 0, QType: PTR	0090	ba ca 27 c2 0b 32 16 f7 67 8a f9 c4 9a ac 50 fa	S.....
Domain Name System (query), QID: 0, QType: PTR	00a0	60 06 74 3c 84 1f 04 6a 56 38 c9 b0 06 0c fa 18	S.....
Domain Name System (query), QID: 0, QType: PTR	00b0	fa b4 93 25 68 1a 0a 03 c1 a5 9a 17 fc 00 91 d6	S.....
Domain Name System (query), QID: 0, QType: PTR	00c0	bc 35 a4 65 97 21 c2 9c 97 c7 04 00 c7 92 1f cd	S.....
Domain Name System (query), QID: 0, QType: PTR	00d0	5d ca 73 ff 14 41 48 59 f7 a8 13 3e 3f 53 55 61	S.....
Domain Name System (query), QID: 0, QType: PTR	00e0	5d 9a 60 7f 2f 64 8b bc 31 58 86 b0 ad 8a 21 30	S.....
Domain Name System (query), QID: 0, QType: PTR	00f0	e5 0a c7 d8 8a 83 99 cc ff cb a2 04 91 ef fa 5c	S.....

No.	Time	Source	Destination	Protocol	Length	Info
116	12.87753600	10.35.226.211	10.35.226.224	DNS	86	Standard query 0x0001 PTR 224.226.35.10.in-addr.arpa
117	12.87753600	10.35.226.211	10.35.226.224	DNS	86	Standard query response 0x0001 No such name PTR 224.226.35.10.in-addr.arpa
118	12.88079500	10.35.226.211	10.35.226.224	DNS	70	Standard query 0x0002 A google.com
119	12.88461300	10.35.226.211	10.35.226.224	DNS	166	Standard query response 0x0002 A google.com A 192.178.158.138 A 192.178.158.139 A 192.178.158.139
120	12.89381300	10.35.226.211	10.35.226.224	DNS	70	Standard query 0x0003 AAAA google.com
121	12.89811400	10.35.226.211	10.35.226.224	DNS	182	Standard query response 0x0003 AAAA google.com AAAA 2404:6800:4013:813::65 AAAA 2404:6800:4013:813::65

Frame 116: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface 0	0000	c2 a8 97 a5 fa 74 98 af 65 f7 1b 15 00 00 45 03	C.....
Ethernet II, Src: Intel_7:7b:15 (98:af:65:f7:1b:15), Dst: Intel_7:7b:15 (98:af:65:f7:1b:15)	0010	00 48 01 b2 00 00 00 11 00 00 0a 23 d2 d3 0a 23	H.....
Internet Protocol Version 4, Src: 10.35.226.211, Dst: 10.35.226.224	0020	e2 a0 c7 6a 00 35 00 34 da 3f 00 01 01 00 00 01	J.....
User Datagram Protocol, Src Port: 53098, Dst Port: 53	0030	00 00 00 00 00 00 03 32 32 34 03 32 32 36 02 33	2.....
Domain Name System (query), QID: 0, QType: A	0040	35 02 31 30 07 69 6e 2d 61 64 64 72 04 61 72 70	5.....
Domain Name System (query), QID: 0, QType: AAAA	0050	61 00 00 0c 00 01	a.....

User Datagram Protocol: Protocol

Packets: 139 - Displayed: 6 (4.3%) - Dropped: 0 (0.0%) Profile: Default

4: DNS Traffic and Query Types in Wireshark

Aim

To capture DNS traffic, identify A, AAAA and MX query types, observe the response, and measure DNS resolution time.

Procedure

1. Open Wireshark.
2. Select Wi-Fi and click Start Capture.
3. Open Command Prompt.
4. Run:

`nslookup google.com`

5. Stop the capture.
6. Apply the filter:

`dns`

Step 1: Identify A Record

In CMD:

`nslookup -type=A google.com`

In Wireshark, look for:

Standard query A google.com

The response will contain an IPv4 address, for example:

google.com → 142.x.x.x

A = IPv4 address

Step 2: Identify AAAA Record

Run:

`nslookup -type=AAAA google.com`

Wireshark should show:

Standard query AAAA google.com

Standard query response AAAA google.com

AAAA = IPv6 address

Your previous screenshot already showed this successfully:

Standard query 0x0003 AAAA google.com

Standard query response 0x0003 AAAA google.com
So you already have a good AAAA output.

Step 3: Identify MX Record

Run:

nslookup -type=MX google.com

Wireshark should show:

Standard query MX google.com

Standard query response MX google.com

MX = Mail Exchange record, which identifies mail servers for the domain.

Step 4: Measure Resolution Time

In Wireshark, find:

DNS Query

and its matching:

DNS Response

For example:

Query 12.893301 sec

Response 12.930511 sec

Therefore:

Resolution time = 12.930511 - 12.893301
= 0.037210 seconds
= 37.21 ms

Your actual time will be different.

Source	Destination	Protocol	Length	Info
18.95.226.211	18.95.226.224	DNS	80	Standard query 80000 PTR 226.226.95.18 IN ANY APOD
18.95.226.224	18.95.226.211	DNS	80	Standard query response 80000 NO SUCH NAME PTR 226.226.95.18 IN ANY APOD
18.95.226.211	18.95.226.224	DNS	78	Standard query 80002 A google.com
18.95.226.224	18.95.226.211	DNS	164	Standard query response 80002 A google.com A 142.250.228.102 A 142.250.228.104 A 142.250.228.106
18.95.226.211	18.95.226.224	DNS	78	Standard query 80003 AAAA google.com
18.95.226.224	18.95.226.211	DNS	182	Standard query response 80003 AAAA google.com AAAA 2001:4860:4004::8d3:1:1 2001:4860:4004::8d3:1:1

Packet, 382 bytes on wire (3056 bits), 382 bytes captured (3056 bits) on 0	0000 90 af 65 f7 10 15 c2 a8 97 a5 fa 00 00 43 00	* * *
Ethernet II, Src: Intel_82:5b:4d:7a:76:76, Dst: Intel_82:5b:4d:7a:76:76	0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
Internet Protocol Version 4, Src: 18.95.226.224, Dst: 18.95.226.211	0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
TCP, Src Port: 55, Dst Port: 50534	0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
System (response)	0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Handwritten signature

CRITERIA FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation